Note

In 1931, H.C. Urey and co-workers noticed that in the shorter wavelength region of the hydrogen spectrum lines, faint companion lines are observed. From the isotope displacement effect (isotope shift), the isotope of the same element can produce slightly different spectral lines. The presence of these faint lines confirmed the existence of isotopes of hydrogen atom (which is named as Deuterium).

On calculating wavelength or wave number difference between the faint and bright spectral lines, atomic mass of deuterium is measured to be twice that of atomic mass of hydrogen atom. Bohr atom model could not explain this isotopic shift. Thus by considering nuclear motion (although the movement of the nucleus is much smaller) into account in the Bohr atom model, the wave number or wavelength difference between the lines produces by the hydrogen atom and deuterium is theoretically calculated which perfectly agreed with the spectroscopic measured values.

The difference between hydrogen atom and deuterium is in the number of neutron. Hydrogen atom contains an electron and a proton, whereas deuterium has an electron, a proton and a neutron.

Excitation energy and excitation potential

The energy required to excite an electron from lower energy state to any higher energy state is known as excitation energy.

The excitation energy for an electron from ground state (n = 1) to first excited state (n = 2) is called first excitation energy.

For hydrogen atom, it is

$E_I = E_2 - E_1 = -3.4\ eV - (-13.6\ eV) = 10.2\ eV$

Similarly, the excitation energy for an electron from ground state (n = 1) to second excited state (n = 3) is called second excitation energy, which is

$E_{II} = E_3 - E_1 = -1.51\ eV - (-13.6\ eV) = 12.1\ eV$

and so on.

Excitation potential is defined as excitation energy per unit charge.

For hydrogen atom, the first excitation state energy is

$$E_I = eV_I$$

First excitation potential for hydrogen atom is,

$$V_I = \frac{1}{e}E_I = 10.2 \text{ volt}$$

Similarly, second excitation potential is,

$V_{II} = \frac{1}{e}E_{II} = 12.1$ volt and so on.

Ionization energy and ionization potential

An atom is said to be ionized when an electron is completely removed from the atom – that is, it reaches the state with energy $E_{n\to\infty}$. **The minimum energy required to remove an electron from an atom in the ground state is known as binding energy or ionization energy.**

For hydrogen atom, the ground state ionization energy is,

$$E_{\text{ionizaation}} = E_\infty - E_1 = 0 - (-13.6eV) = 13.6eV$$

When an electron is in n^{th} state of an atom, the energy required to remove an electron from that state – that is, the corresponding ionization energy is

$$E_{\text{ionization}} = E_\infty - E_n = 0 - \left(-\frac{13.6}{n^2}Z^2\ eV\right) = \frac{13.6}{n^2}Z^2\ eV$$

At normal room temperature, the electron in a hydrogen atom (Z=1) spends most of its time in the ground state. **The amount of**

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